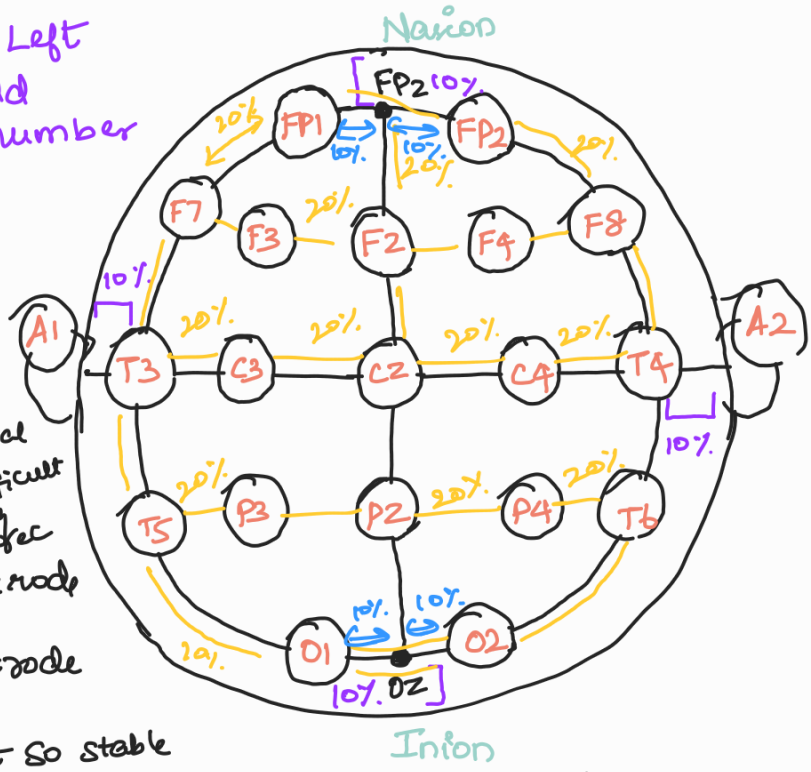
13/3/26

Standardized Placement of electrode.

Left
odd number

Right
even number



Spacing → 20%
spacing → 10%

↓

according to 10 - 20% international Standard system for placing electrode.

Stable signal but difficult for long sec ↑

1. wet electrode
2. Dry electrode
   ↳ not so stable
   not required gel, can be used for long term.

Things to note :

① Impedance of electrode needs to be below 5 KΩ to transmit clear signal, if not it may provide signals with lots of noise, which leads to low quality EEG signal.

② When turning ON the electrode, the display of green light in electrode represent that they have right impedance. If it shows red then we need to reduce the impedance.

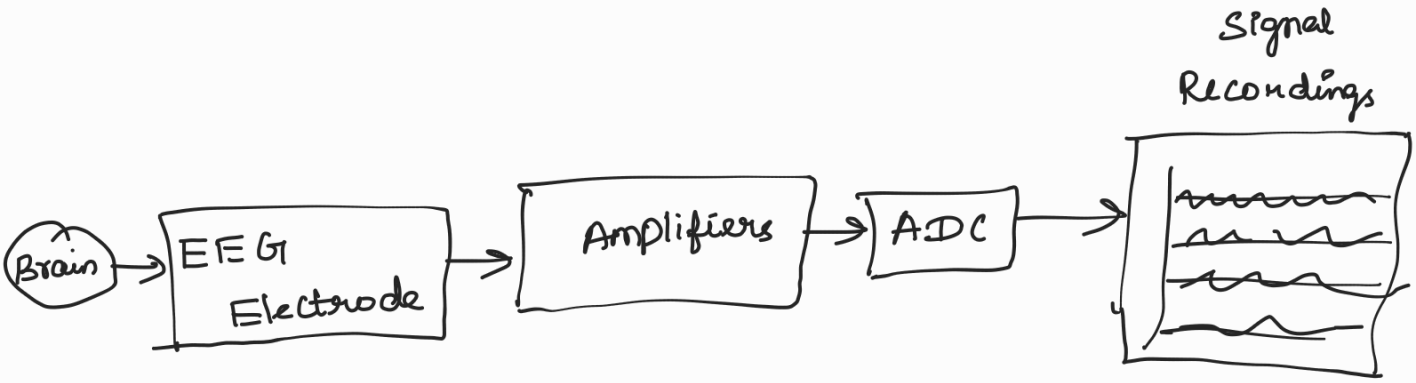
③ Methods to reduce Impedance :-

- Application of conductive gel.
- Cleaning the scalp
- Ensure tight contact b/w electrode & skin.

Reference electrode

1. CZ
2. Linked ears
3. Linked mastoids
4. Ipsilateral ear
5. contralateral ear
6. Nasion.

It is important because the EEG measures the potential b/w two electrodes.



Common file format for EEG data storage.

1. EDF
2. BDF
3. CSV

Includes details like

1. Sampling rate
2. Time series data
3. Channel information
4. Other device details

1. MNE libraby is used to read the EDF file for processing bio signal data, specifically EEG.

```
import mne
import numpy as np
```